GEOMETRICAL AND WAVE OPTICS

Track 5 — Diverging Lenses: The Peephole

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Problem Statement

A diverging lens of focal length **f' = −10 cm** (optical power **P = −10 D**) is placed in the peephole of a door. A guard of height **AB = 1.80 m** stands at a distance **OA = −50 cm** in front of the lens.

Answer the following questions:

Q1 — Calculate the optical power P of the lens.

Q2 — Determine the position OA' of the image formed by the lens.

Q3 — Calculate the lateral magnification M.

Q4 — Determine the height A'B' of the image.

Q5 — Explain why a diverging lens widens the field of view of a peephole.

Q1 — Optical Power

Definition

The optical power of a lens is defined as the inverse of its focal length expressed in metres:

```
P = 1 / f'
```

Calculation

Converting the focal length to metres:

```
f' = −10 cm = −0.10 m
```

Applying the definition:

```
P = 1 / (−0.10) = −10 D
```

P = −10 Diopters

A negative power confirms that this is a diverging lens. Incident rays are refracted outward as if they originated from a virtual focal point F located 10 cm in front of the lens on the same side as the incoming light.

Q2 — Image Position (OA')

Thin Lens Equation

The Descartes thin lens formula (conjugate relation) is:

```
1/OA' − 1/OA = 1/f'
```

Calculation

Rearranging to isolate 1/OA':

```
1/OA' = 1/f' + 1/OA
```

Substituting the known values (f' = −10 cm, OA = −50 cm):

```
1/OA' = 1/(−10) + 1/(−50)
1/OA' = −5/50 + (−1/50) = −6/50
OA' = −50/6 ≈ −8.33 cm
```

OA' = −8.33 cm

Since OA' < 0, the image forms on the same side as the object (in front of the lens). This is a virtual image — it cannot be projected onto a screen, but is clearly perceived by the eye looking through the lens.

Q3 — Lateral Magnification (M)

Definition

The lateral magnification is defined as:

```
M = OA' / OA
```

Calculation

Substituting OA' = −8.33 cm and OA = −50 cm:

```
M = (−8.33) / (−50)
```

M = +0.167

Interpretation of the sign and magnitude:

M > 0 → The image is **erect (upright)**: it has the same orientation as the object.

|M| < 1 → The image is **reduced (smaller)**: the guard appears approximately 6× smaller than his real height.

Q4 — Image Height (A'B')

Definition

The image height is related to the object height by:

```
A'B' = M × AB
```

Calculation

Substituting M = +0.167 and AB = 1.80 m:

```
A'B' = 0.167 × 1.80
```

A'B' = 0.30 m = 30 cm

The guard who is 1.80 m tall in reality appears only 30 cm tall through the diverging lens — a reduction by a factor of approximately 6×. This reduction is precisely what enables the widening of the field of view (see Q5).

Q5 — Why Does a Diverging Lens Widen the Field of View?

Principle

Without any lens, a bare peephole (a simple aperture) only allows rays that travel nearly parallel to the optical axis to pass through. Wide-angle rays from objects at the periphery of the scene cannot enter the aperture and are therefore lost — the field of view is extremely narrow.

When a diverging lens is placed in the peephole, it **bends incoming wide-angle rays inward** toward the optical axis, allowing them to pass through the aperture. The lens then forms a **virtual, erect, and reduced** image of the entire wide scene. Because the image is compressed (M = +0.167, approximately 6× smaller), a much wider angular region of the scene fits within the aperture and reaches the observer's eye.

Key Properties of the Diverging Lens Image

1. Always Virtual: OA' < 0 → image on same side as object, cannot be projected.

2. Always Erect: M > 0 → image has same orientation as object.

3. Always Reduced: |M| < 1 → image is smaller than object.

4. Always between F and the lens: The image forms closer to the lens than the real focal point.

> *Note: The exact field-of-view angle depends on the aperture diameter and lens geometry and is not derived from the given data. Typical commercial peephole lenses provide a field of view of approximately 160° compared to only a few degrees for a bare hole — illustrating the dramatic practical benefit of the diverging lens.*

Summary of Results

Question	Formula	Substitution	Result	Interpretation
Q1 — Power P	**P = 1 / f'**	1 / (−0.10 m)	**P = −10 D**	**Diverging lens**
Q2 — Image OA'	**1/OA' − 1/OA = 1/f'**	−1/10 + (−1/50)	**OA' = −8.33 cm**	**Virtual image**
Q3 — Magnif. M	**M = OA' / OA**	(−8.33) / (−50)	**M = +0.167**	**Erect, reduced**
Q4 — Height A'B'	**A'B' = M × AB**	0.167 × 1.80 m	**A'B' = 30 cm**	**6× smaller**
Q5 — FOV	**Angular compression**	Wide → Narrow	**Widened FOV**	**Qualitative result**

Conclusion

The diverging lens inserted in the peephole has an optical power of **P = −10 D**. It forms a virtual, erect, and reduced image of the guard at **OA' = −8.33 cm** from the lens, with a lateral magnification of **M = +0.167**. The guard's apparent height through the lens is only **A'B' = 30 cm** instead of 1.80 m. This size reduction is the physical mechanism that widens the field of view of the peephole, allowing observation of a much broader scene from inside the door without being seen.